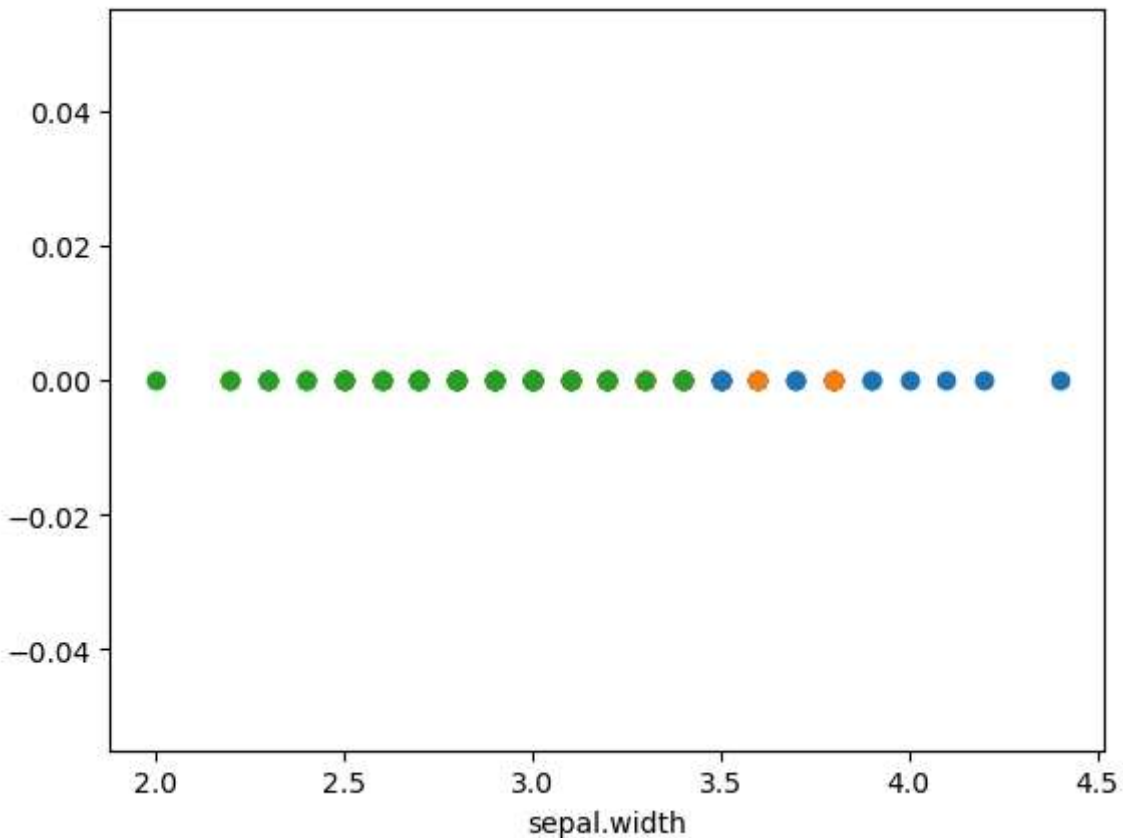
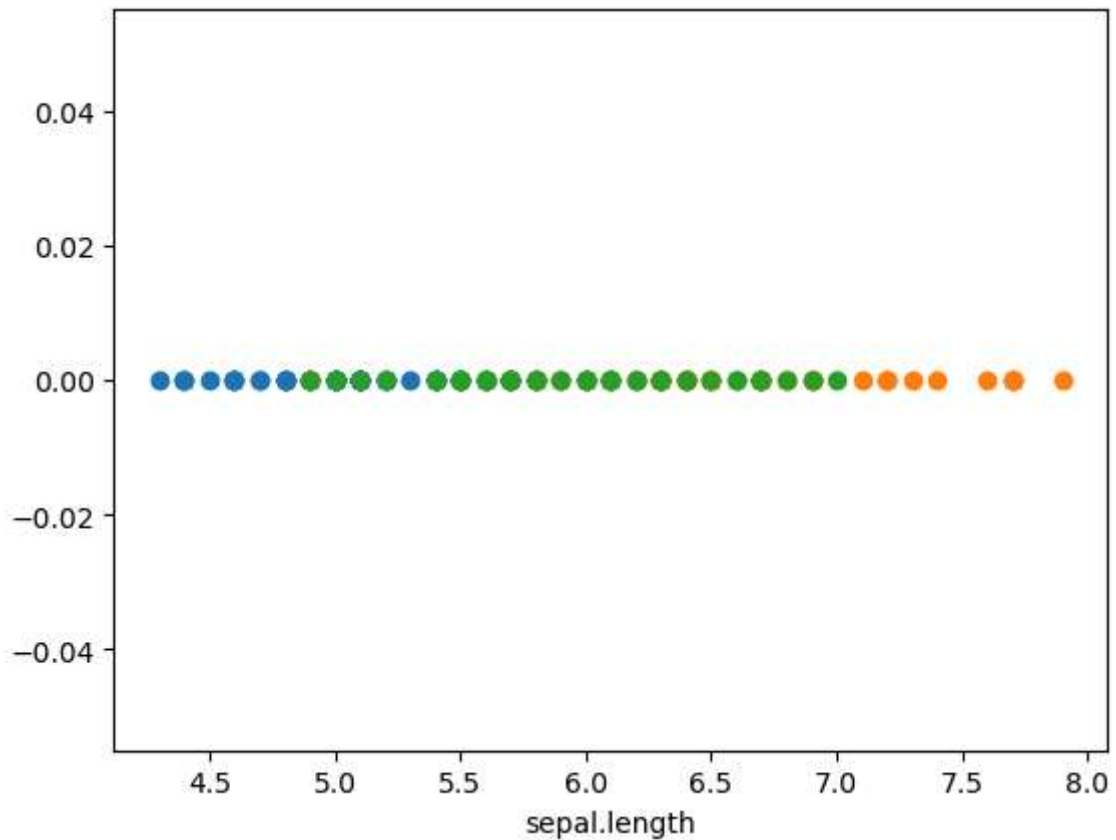


```
In [6]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
df = pd.read_csv('iris.csv')
df.head(150)
df.shape
df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['sepal.width'],
            np.zeros_like(df_Setosa['sepal.width']))
plt.scatter(df_Virginica['sepal.width'],
            np.zeros_like(df_Virginica['sepal.width']))
plt.scatter(df_Versicolor['sepal.width'],
            np.zeros_like(df_Versicolor['sepal.width']))
plt.xlabel('sepal.width')
plt.show()
```

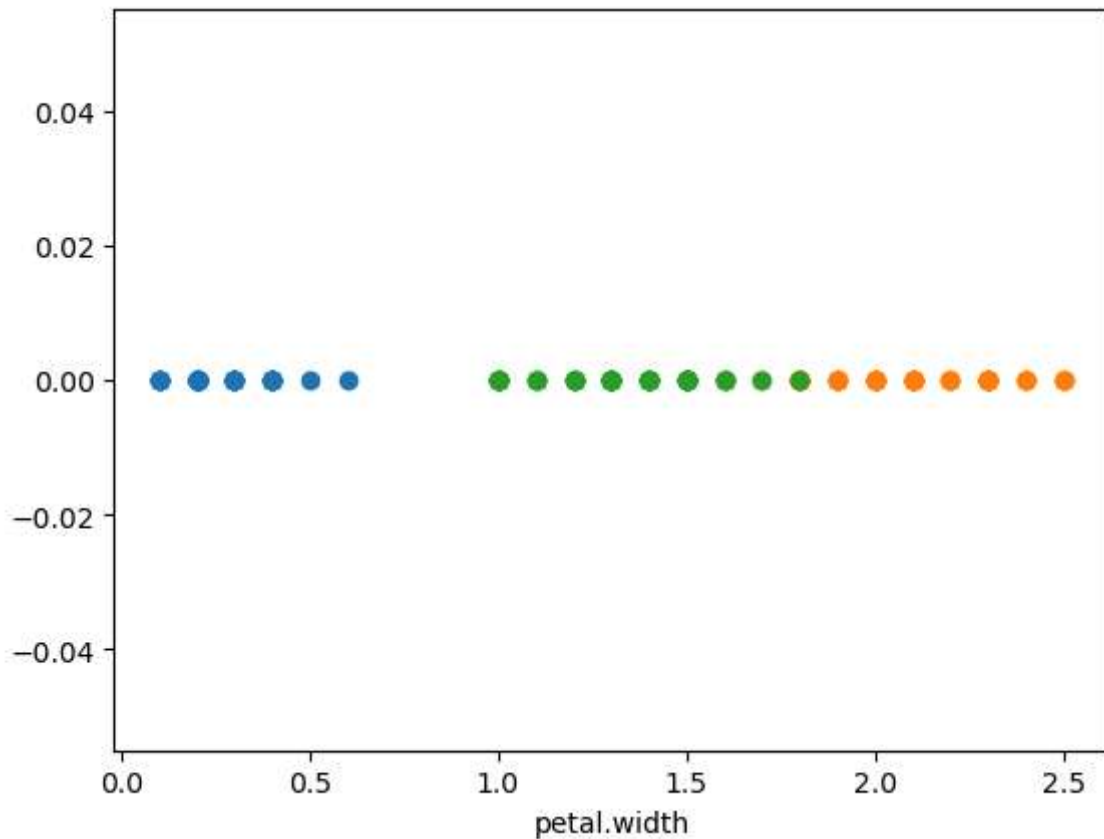


```
In [7]: df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['sepal.length'],
            np.zeros_like(df_Setosa['sepal.length']))
plt.scatter(df_Virginica['sepal.length'],
            np.zeros_like(df_Virginica['sepal.length']))
```

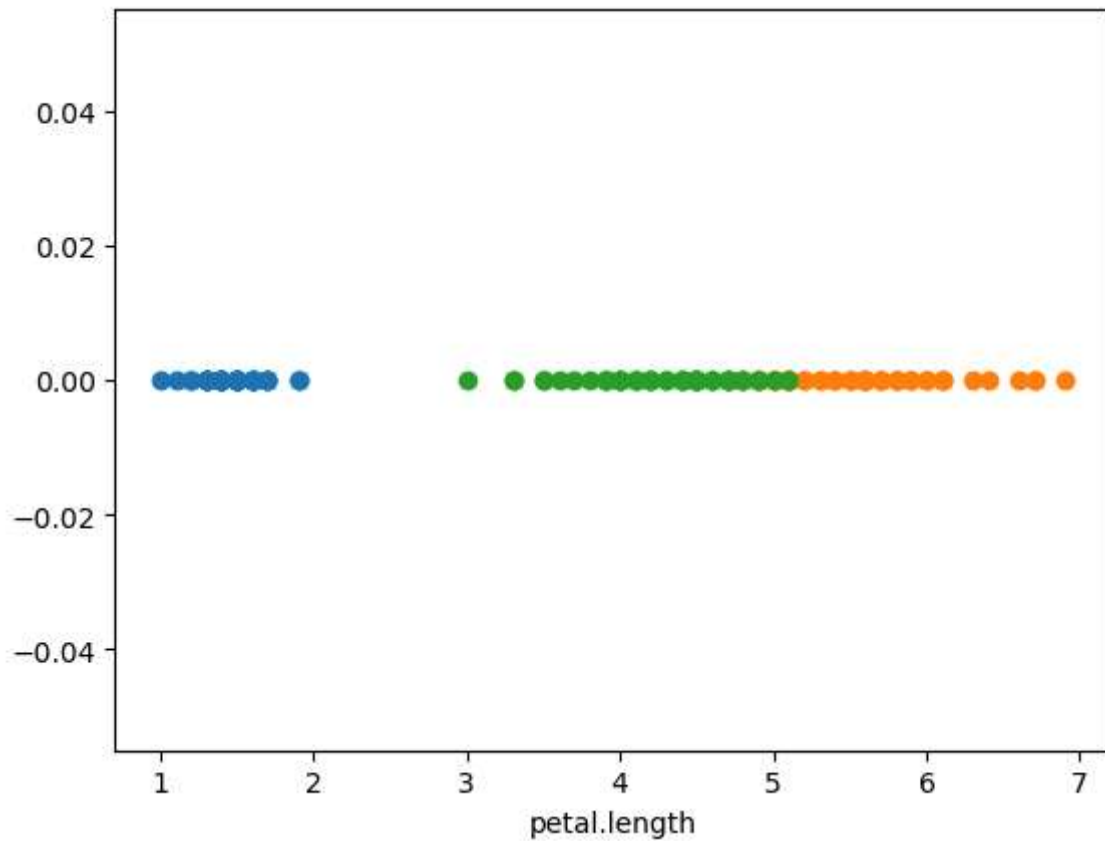
```
plt.scatter(df_Versicolor['sepal.length'],
            np.zeros_like(df_Versicolor['sepal.length']))
plt.xlabel('sepal.length')
plt.show()
```



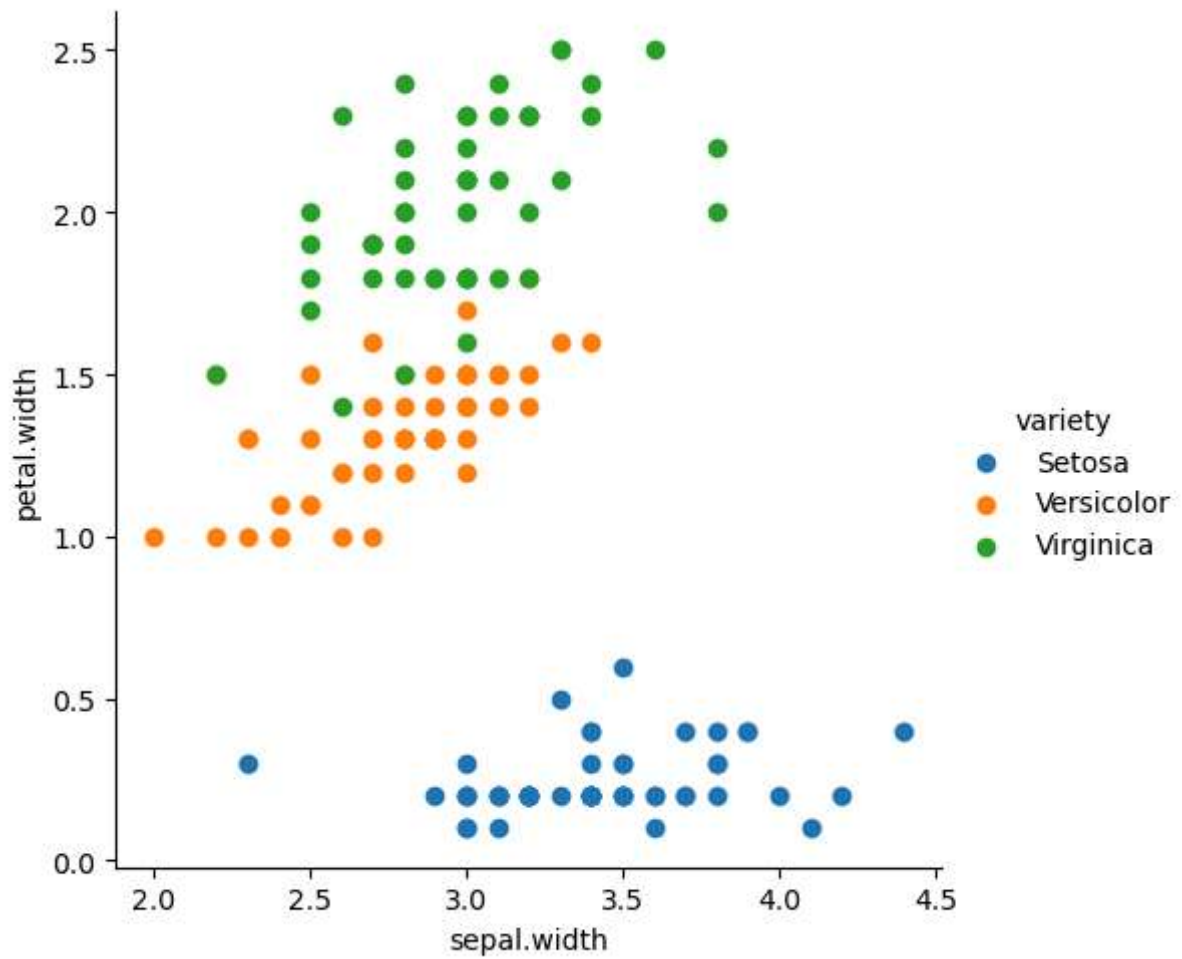
```
In [8]: df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['petal.width'],
            np.zeros_like(df_Setosa['petal.width']))
plt.scatter(df_Virginica['petal.width'],
            np.zeros_like(df_Virginica['petal.width']))
plt.scatter(df_Versicolor['petal.width'],
            np.zeros_like(df_Versicolor['petal.width']))
plt.xlabel('petal.width')
plt.show()
```



```
In [9]: df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['petal.length'],
            np.zeros_like(df_Setosa['petal.length']))
plt.scatter(df_Virginica['petal.length'],
            np.zeros_like(df_Virginica['petal.length']))
plt.scatter(df_Versicolor['petal.length'],
            np.zeros_like(df_Versicolor['petal.length']))
plt.xlabel('petal.length')
plt.show()
```



```
In [11]: sns.FacetGrid(df, hue='variety', height=5)\
        .map(plt.scatter, "sepal.width", "petal.width")\
        .add_legend()\
        plt.show()
```



```
In [14]: sns.FacetGrid(df, hue='variety', height=5)\n        .map(plt.scatter, "sepal.length", "petal.length")\n        .add_legend()\n\nplt.show()
```

